



Society of Petroleum Engineers

SPE-229682-MS

Future of Upstream in the Era of Agentic AI

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

The upstream oil and gas (O&G) sector is at a critical inflection point. While past decades of automation and digitalization have significantly optimized workflows, they still relied heavily on human oversight. Today, the emergence of Agentic AI marks a transformative shift—enabling intelligent agents to independently execute complex tasks, make context-aware decisions, and dynamically adapt to real-time operational data, significantly reducing the need for constant human involvement.

Unlike earlier incremental advances, Agentic AI represents a fundamental disruption. These autonomous systems integrate technologies such as large language models (LLMs) with decades of domain-specific data, enabling capabilities far beyond traditional automation. This paper explores the transformative potential of Agentic AI through the lens of ENERGYai—a novel AI framework purpose-built for upstream applications.

We examine how Agentic AI is being implemented across critical upstream workflows—from seismic interpretation and reservoir modeling to drilling optimization and production management. ENERGYai, for instance, has delivered seismic analysis up to 80% faster, increased production efficiency by as much as 15%, and reduced unplanned downtime through proactive asset monitoring and intervention. Interestingly, digitally native junior professionals are seeing disproportionate productivity gains, as Agentic AI augments their domain knowledge with real-time insights and decision support. This shift is not just about technology—it's about workforce transformation. However, key challenges remain ensuring data quality, implementing robust QA/QC processes, and, crucially, building trust in autonomous decision-making systems.

Change management is essential. Successful adoption requires not only technical integration but also cultural readiness. Organizations that empower agile, AI-literate talent, embrace open data standards, and shift away from restrictive legacy systems and licensing models will be best positioned to lead this next era of upstream innovation.

In conclusion, Agentic AI redefines how upstream operations are conceived and executed. It offers a step-change in efficiency, responsiveness, and innovation. Companies that move decisively to adopt these technologies will unlock significant competitive advantages—while those that resist risk being left behind in a rapidly evolving, AI-driven industry landscape.

Introduction: From Automation to Agentic AI in Upstream

Upstream O&G's Technological Evolution. The upstream sector (exploration and production) has historically been heavily reliant on human expertise – from geoscientists interpreting seismic surveys to engineers managing reservoirs. In the last two decades, automation and digitalization have gradually improved efficiency. Remote sensors, PLCs, and digital oilfield initiatives began to replace some manual monitoring, and big-data analytics enhanced predictive capabilities (e.g. forecasting equipment failures or optimizing production rates)[3][4]. These advances delivered clear benefits – faster data processing, reduced human error, and improved safety – yet they stopped short of replacing human judgment. Most digital workflows still required significant manual oversight and intervention, and many companies remained burdened by siloed data in legacy systems and monolithic software with complex licensing models[1][5]. In fact, the industry became commoditized for its reliance on proprietary petrotechnical applications that locked data into vendor-specific formats, creating "information islands" and persistent inefficiencies[5][6].

Agentic AI. Recent breakthroughs in AI are now enabling a more profound transformation. *Agentic AI* refers to autonomous AI systems – "agents" – that can perceive their environment, reason and plan actions, and execute decisions to achieve specific objectives without needing step-by-step human guidance. These agents are adaptive and context-aware, collaborating with humans and other systems in dynamic operational environments[2]. In upstream O&G, Agentic AI systems go far beyond static automation scripts or standalone predictive models. They integrate advanced machine learning (incl. computer vision, deep and reinforcement learning), real-time sensor data streams, and even LLMs to analyze complex situations, support making decisions, and carry out tasks in a goal-directed manner. Crucially, they **learn continuously** from new data and outcomes, improving their performance over time – a huge contrast to traditional software that follows rule-based approach.

A Paradigm Shift. The advent of agentic AI represents a paradigm shift rather than a gradual step. Previous technological initiatives often automated *parts* of a workflow (e.g., using software to map a seismic attribute or running a predictive maintenance alert), but still left final interpretations and decision-making to humans. Now, AI agents can handle entire decision loops autonomously. For instance, an AI drilling agent might monitor downhole sensor data in real time, adjust drilling parameters on the fly, predict a dangerous pressure influx, and execute preventative measures – all autonomously and within seconds, without awaiting human approval[7][8]. This ability to act in real time based on environmental feedback is why industry leaders are calling agentic AI "the next frontier in industrial automation". In an IBM survey, 29% of upstream organizations indicated they are already piloting agentic AI optimization (such as dynamically adjusting mud weights to prevent blowouts), recognizing the potential of these autonomous agents to transform decision-making and operational execution[9].

ENERGYai – A First-of-its-Kind Case

A prominent example of this shift is **ENERGYai**, developed by AIQ in partnership with ADNOC. Launched as the world's first agentic AI solution tailored for the energy sector, ENERGYai integrates a massive 70-billion-parameter LLM with over 50 years of ADNOC's proprietary upstream data[10]. The system comprises multiple specialized AI agents trained on specific tasks across the upstream value chain. In a 90-day proof-of-concept trial, ENERGYai demonstrated dramatic improvements in both speed and accuracy of key upstream activities. Notably, it delivered a **70% improvement in accuracy** on major seismic interpretation tasks and produced "rapid, precise and detailed" seismic analysis results that provided actionable insights for production optimization[11][12]. These AI-generated insights were delivered in natural language, allowing engineers to interact with the system as they would with a knowledgeable PhD-level expert[13]. Such results confirm the transformational potential of agentic AI – validating that these

agents can match or exceed human expert performance in complex geological interpretation, while doing so faster and at greater scale[14].

In summary, the upstream industry is moving from an era where humans use tools to an era where autonomous tools make decisions. The sections that follow detail how agentic AI is implemented in upstream operations, the benefits and real-world results observed, and the implications for the workforce and industry at large.

Implementing Agentic AI in Upstream Operations

Adopting agentic AI in upstream is not a simple IT upgrade – it is a comprehensive transformation of workflows, data infrastructure, and organizational processes. **ENERGYai's** deployment provides a useful blueprint, consisting of several strategic elements:

Unified Data Integration

The foundation of any AI-driven system is data. Upstream companies typically possess decades of historical data – seismic surveys, well logs, production records, maintenance logs, geological models, etc. However, these are often scattered across disparate databases and file systems, or locked within vendor-specific applications. The first step is consolidating and standardizing this data into a unified platform. Initiatives like the *Open Subsurface Data Universe (OSDU)* have been pivotal in this regard. OSDU is an open industry data platform that aims to eliminate silos by decoupling data from applications and using common formats. By aggregating data under OSDU principles, **ENERGYai** enabled its AI agents to readily access and cross-analyze previously isolated datasets (for example, correlating seismic data with production and well intervention data). A standardized, cloud-backed data lake not only improves data quality and consistency but also reduces dependence on proprietary data stores, thus **avoiding vendor lock-in** and enabling more flexible analytics[5][6]. In practice, ADNOC loaded petabytes of subsurface and operational data into **ENERGYai's** backend, ensuring the AI agents could draw on a rich, contextual knowledge base spanning the company's entire operational history.

Specialized AI Agents for Key Workflows

Rather than a monolithic AI, agentic systems consist of multiple specialized agents, each designed for a critical domain task. **ENERGYai**, e.g., is working on **fully operational AI agents** covering various subsurface operations. These can be thought of as a suite of "digital specialists" working in tandem. Examples include:

- **Seismic Interpretation Agent (an "AI Geologist"):** Ingests seismic volumes and well data to automatically identify geological features – faults, horizons, stratigraphic facies – and propose prospects. Deep learning models in this realm have achieved fault detection accuracy exceeding 90%, outperforming traditional manual interpretation (typically 70–80% accuracy)[14]. By using such models, the AI Geologist agent can rapidly map subsurface structures with confidence levels comparable to experienced interpreters. In **ENERGYai's** trial, an AI seismic agent identified fault lines and horizons with **over 70% accuracy** (even at its early stage), already beating labor-intensive human methods in speed and consistency[14]. It also reduced interpretation *time* by an estimated 80% – tasks that once took geoscientists weeks of manual work were completed in hours. This dramatic acceleration (without proportionate loss of accuracy) frees human experts to focus on higher-level geological reasoning and prospect validation, rather than painstaking line-by-line interpretations.
- **Reservoir Engineering and Optimization Agents:** These include agents like an "AI Petrophysicist" to analyze well logs and core data, a **Reservoir Monitoring Agent** to continuously track reservoir performance, and production optimization agents. They autonomously perform

tasks such as quality-controlling incoming field data, recalibrating reservoir models, and running simulations to recommend optimum well settings (e.g., choke adjustments, gas injection rates). By leveraging machine learning on historical production and pressure data, such agents can spot anomalies or suboptimal conditions far faster than periodic human analysis. For example, if water cut in a producer well starts rising or a certain reservoir zone shows unexpected pressure behavior, the AI flags it and might autonomously adjust injection patterns or suggest a workover. In ENERGYai's deployment, junior reservoir engineers were able to *delegate* routine data QC and surveillance tasks to these agents. The result was a step-change in efficiency: data cleaning and validation that used to take weeks of human effort were done continuously in real-time, leading to more trustworthy data feeds and quicker decision cycles. Importantly, these AI agents' recommendations led to proactive reservoir management decisions that increased forecast accuracy and production tuning. External studies echo such gains – for instance, ExxonMobil reported a **10% increase in oil production** in certain fields by using machine learning to optimize waterflood management (with some wells seeing up to 20% gains). Broadly, AI-driven reservoir optimization can boost production efficiency on the order of 5–15% in many cases, by extracting insights from data that humans might miss.

- Drilling and Operations Agents:** These include autonomous drilling rig agents and predictive maintenance agents. A drilling agent can monitor data from drill bits, mud logs, and rig sensors to make instantaneous adjustments. For example, BP deployed machine learning on its Gulf of Mexico rigs and achieved a **20% reduction in drilling time** while cutting costs by millions, by allowing AI to optimize bit parameters and drilling path in real-time[8]. Similarly, Devon Energy's AI-steered rigs can navigate complex shale formations with precision previously unattainable, reducing the need for costly directional drilling corrections. In addition to speed, **safety and risk management** improve: AI can predict kicks or equipment failures before they happen. A *predictive maintenance agent* analyzes sensor data from pumps, engines, and blowout preventers to detect early signs of trouble. Shell, for instance, reduced unscheduled downtime by 20% and lowered maintenance costs 15% by adopting AI-driven predictive maintenance on its rigs. In upstream, such AI agents can automatically schedule preventative maintenance or adjust operating parameters to avoid failures, thereby **cutting downtime** and minimizing HSE (health, safety, environment) incidents.
- QA/QC and Data Prep Agents:** A less glamorous but crucial role for AI agents is in data quality assurance and preparation. Upstream workflows involve enormous data streams (seismic traces, log depths, production volumes, etc.), which historically involve tedious manual cleaning and formatting. Agentic AI can automate much of this. In the ENERGYai trial, an agent continuously scanned incoming operational data for errors, outliers, and format issues – *vastly improving data reliability and usability* by standardizing units, filling gaps, and flagging implausible entries. This ensured that downstream analysis (whether by humans or other AI agents) was based on high-quality data, thereby improving the overall accuracy of decisions. Such continuous QA/QC is essential in high-stakes contexts, as poor data can otherwise lead AI astray. By handling data grunt-work 24/7, the agents again free human experts to concentrate on interpretation and strategy.

Continuous Autonomous Learning

Agentic AI systems are not "fire-and-forget" software; they are designed to learn and adapt. Through techniques like reinforcement learning and feedback loops, the AI agents iteratively improve their models as new data arrives. For example, a seismic interpretation agent might initially be trained on historical datasets, but as it processes new surveys and receives validation from drilled well results, it updates its neural network to better predict geology. ENERGYai's agents employ **real-time data feedback** to self-improve – akin to having an ever-learning junior analyst who remembers every outcome. This continuous

learning was evident in anomaly detection during the trial: as the AI monitored reservoir performance, whenever its predictions deviated from observed results, it treated it as new training data to refine future predictions[12]. Similarly, the drilling agent's models get better with each well drilled, learning the subtle patterns leading up to equipment wear or a drilling problem. The **feedback loop** capability is critical in upstream because each reservoir and operation is unique; an AI must adapt to site-specific conditions. Modern digital oilfield architectures facilitate this by streaming sensor and production data into cloud AI platforms **continuously**. Over time, an autonomous agent that is allowed to learn can surpass even experienced humans in consistency, as it accumulates knowledge of edge cases and rarely occurring events that a person might not encounter often in their career.

Autonomous Operational Oversight

As agents take on more decision-making, a new approach to oversight is required. Instead of humans supervising every decision, the AI system itself can monitor operations at a meta-level and flag exceptions. Think of it as *AI supervising AI*, with humans managing by exception. ENERGYai's framework includes real-time performance monitoring dashboards where the output of each agent is tracked against key metrics (accuracy, confidence level, KPIs like drilling rate or production variance). If an agent encounters a scenario outside its trained experience or if conflicting objectives arise, it can automatically escalate to a human or to another supervisory agent. For example, a **safety oversight agent** might watch all operational suggestions and intervene if any recommendation might breach safety constraints (like a drilling parameter outside safe bounds). This approach significantly **decreases the need for routine manual supervision** yet retains human control for unusual or high-risk decisions. In practice, operators in the field would receive concise alerts or advice from AI (e.g. *"Predicted equipment failure in 3 days; maintenance scheduled"*) and only need to step in if they disagree or if the AI signals uncertainty. Such human-on-the-loop design builds trust – engineers know they haven't entirely ceded control, and they can audit the AI's rationale delivered in natural language[13]. Indeed, ADNOC engineers reported that interacting with ENERGYai's insights via an intuitive interface made it easier to accept and act on the AI's recommendations[28]. Over time, as confidence in autonomous operations grows, the oversight can be further relaxed or focused on only the most critical decisions.

Change Management and Workflow Reengineering

Implementing agentic AI is as much an organizational challenge as a technical one. Traditional upstream workflows and job roles must be reimaged. Simply deploying AI tools without altering business processes will lead to suboptimal outcomes or even resistance. *Change management* is therefore a pivotal element of success. Companies must foster a culture that embraces AI-driven workflows, which means addressing employee concerns, retraining staff, and redefining roles. In upstream terms, this means clearly communicating to geoscientists and engineers **how agentic AI will augment (not replace)** their roles, demonstrating early "quick wins" where AI support made projects faster or safer, and providing training so staff can effectively work *with* AI agents. For instance, an exploration team might be trained to treat the AI seismic interpreter as a colleague – learning how to query its results, provide feedback on misses, and incorporate its analysis into their prospect evaluation process. Companies like ADNOC have paired subject matter experts with data scientists and AI specialists during pilot phases, ensuring that domain knowledge is transferred to the AI and that the end-users trust the outcomes.

Crucially, adopting agentic AI often entails **reengineering workflows** from end to end. It's not just automating one step, but potentially changing the sequence and responsibilities. **Change management** also means tackling the cultural shift: encouraging experienced staff to adopt a "teach and trust the AI" mindset and encouraging cross-generational collaboration so that tech-savvy junior staff and seasoned experts can learn from each other.

In implementing these steps, ENERGYai and similar initiatives underscore that success comes from a harmonious integration of technology, data, and people. Next, we examine one particularly provocative implication of this transformation – the differing impact on younger versus older professionals in upstream – followed by a look at the tangible performance outcomes achieved with agentic AI.

Workforce Dynamics: Empowering the Digital-Native Professionals

One striking observation from early deployments of agentic AI in upstream is the **disproportionate advantage it provides to younger, more technologically agile professionals**. The upstream sector has a well-documented generation gap, with an aging workforce on one side and a wave of digitally-native entrants on the other. As agentic AI systems roll out, junior subject matter experts (SMEs) – typically in their 20s and 30s, comfortable with modern software and open to new workflows – have been quick to embrace and excel with these tools. In contrast, some senior SMEs, though rich in experience, find it challenging to unlearn entrenched methods or fully trust AI-driven processes. This generational difference has tangible implications for companies managing the transition.

Adaptability of Junior Staff. Younger engineers and geoscientists have grown up in a digital world. They tend to be more adaptable and receptive to cutting-edge tools, treating AI as a natural extension of their toolkit. Early-career workers are indeed more positive about AI – a higher proportion of them believe that AI enhances their work quality, job satisfaction, and growth opportunities compared to their more tenured colleagues. AI frees them from tedious tasks, allowing them to focus on more complex problem-solving that was traditionally the domain of senior staff. We see this dynamic in upstream teams adopting ENERGYai. For example, **junior geoscientists** were quick to adapt the "AI Geologist" agent into their interpretation workflow. Rather than spending days manually picking seismic horizons, they let the AI agent do the first pass, then they focused on validating and refining the AI's picks. Within weeks, these young professionals were achieving interpretation turnaround times unimaginable before – sometimes delivering initial prospect analyses in *one-fifth* the time previously needed. They effectively **blended their growing domain knowledge with AI-generated insights**, resulting in a synergy where the whole was greater than the sum of parts.

Challenges for Senior Staff. On the other hand, senior SMEs – often with decades of ingrained habits – may initially be more skeptical of AI recommendations or simply less comfortable interfacing with new digital tools. There can be an instinct to trust personal judgment over an algorithm, especially in an industry where experience has long been the gold standard. Some veteran engineers worry that if AI handles many foundational tasks, the next generation might miss out on learning opportunities that come from doing those tasks manually. There can also be anxiety about job displacement or erosion of authority. It's worth noting, however, that many experienced professionals who do engage with the AI soon recognize its value as a tireless assistant. The key is involving them in training the AI (so their expertise is embodied in it) and showing respect for their oversight role.

New Roles and Growth Opportunities. As routine tasks become agent-driven, the human roles evolve. We are seeing the emergence of what might be called *AI-augmented specialists*. For example, a reservoir engineer's job might shift from manually running simulation cases to curating and validating the scenarios proposed by an AI agent, then focusing on field development strategy and cross-discipline coordination less number-crunching, more decision-making. Early adopters of AI among the younger cohort find they can take on **bigger responsibilities faster**, because the AI serves as a force-multiplier for their skills. This can flatten traditional hierarchies to some extent: a junior engineer with AI support can produce analysis on par with a mid-level engineer, potentially accelerating their career progression. As one young professional remarked, *AI "gave me a few extra hours in the day to work on challenging problems, which used to be*

spent on tedious prep". Management should harness this enthusiasm – these digitally native employees can become **AI champions** internally, helping train colleagues and refine the AI tools.

From a leadership perspective, leveraging the strengths of each generation will be key. Younger staff drive innovation and quick adoption; senior staff provide critical oversight, context, and risk management. Those companies that empower their junior workforce with AI (while retraining senior staff to utilize and supervise AI effectively) will cultivate a more dynamic and future-proof talent pool. Notably, organizations may need to update their talent development models: instead of the traditional ladder where expertise comes with years of manual work, future career paths might involve faster assumption of decision roles, with AI handling the grunt work. This can be exciting but also requires careful mentoring to ensure deep domain expertise is still built over time (by exposing young engineers to a wider variety of scenarios via AI).

Expected Tangible Benefits and Early Results

Although agentic AI in upstream is still an emerging trend, early deployments like ENERGYai have yielded **substantial improvements** in key performance metrics. These demonstrate the technology's game-changing potential when scaled. We summarize the major benefits reported:

- **Seismic Interpretation Efficiency:** The most dramatic gains have been in subsurface data analysis. AI seismic agents have slashed interpretation cycle times by as much as 80%, while achieving accuracy levels on par with or even exceeding human experts. In ENERGYai's trial, seismic analysis that once took geoscientists months was completed in days, accelerating ADNOC's upstream exploration pace. Accuracy in identifying critical features (faults, horizons, direct hydrocarbon indicators) improved markedly – the trial noted a **70% improvement in accuracy** on major seismic interpretation tasks[12]. This is corroborated by other studies where deep learning models reached over 90% accuracy in fault detection, versus ~70% by manual methods[14]. While the AI might not perfectly interpret every geological nuance, it provides a highly reliable first pass that narrows down targets quickly. The net effect is faster prospect generation, more precise well placement, and fewer missed opportunities. One ENERGYai statistic indicated that with AI assistance, interpreters evaluated 4 times more seismic sections per week than before, without sacrificing quality. In an industry where time equals money (rig rates, license deadlines, etc.), this **speed-up can save millions and improve competitiveness** in securing reserves.
- **Improved Reservoir Management and Production Optimization:** By continuously analyzing production data and updating reservoir models, AI agents help unlock additional hydrocarbons and optimize recovery strategies. ENERGYai's reservoir-focused agents increased predictive accuracy for reservoir performance, contributing to **up to 15% greater production efficiency** in pilot areas (e.g. by suggesting better well choke settings and lift optimizations). This aligns with field case studies: for instance, a reinforcement learning approach to an enhanced oil recovery process achieved a **15% increase in oil recovery** compared to traditional methods. Similarly, one of the applications of AI to waterflood optimization led to about a 10% production uplift overall. Such improvements can translate to significant volumes in large fields – effectively extracting more O&G with the same assets. Additionally, AI-driven reservoir surveillance can anticipate problems like water breakthrough or pressure depletion earlier. Operators can then proactively intervene (e.g. shifting injector patterns, doing infill drilling) to mitigate declines, thereby extending field life. In the long term, these efficiencies might also allow companies to trim capital expenditures – squeezing more out of existing wells rather than drilling new ones, which is especially valuable in a low-carbon, capital-disciplined future.
- **Reduced Operational Risks and Downtime:** Safety and uptime are paramount in upstream operations, and here too agentic AI has shown notable benefits. **Predictive maintenance and drilling optimization agents** have cut unplanned downtime and helped avoid incidents. Fewer

surprises during drilling not only save costs but also significantly reduce HSE risk (each avoided kick or blowout is a potential life and environment saver). Moreover, autonomous monitoring agents improve day-to-day safety: for example, by continuously scanning sensor data, an AI can detect abnormal gas levels or pressure buildups and initiate shutdowns or alerts within seconds. The reliability of such AI detection can surpass human vigilance, especially at facilities with thousands of signals. Another risk reduction is in logistics and asset integrity – pipeline monitoring agents, like those piloted in midstream, use sensor and satellite data to spot leaks or weaknesses early[9]. By catching these issues proactively, companies prevent accidents and environmental harm *and* avoid costly downtime from pipeline outages. Overall, the introduction of AI-driven predictive systems has improved operational continuity and safety metrics: some operators report **incident rates dropping** as the AI helps enforce safety checks and optimize conditions (for example, maintaining equipment within ideal operating envelopes to avoid fatigue).

- **Cost Savings and Efficiency Gains:** Many of the above improvements also drive cost reduction. Faster interpretations and optimized drilling mean lower exploration cost per barrel found. Better production efficiency and less downtime yield higher output for the same operating cost. Additionally, agentic AI can reduce labor and software costs in the long run. There is also the potential to **escape traditional software vendor lock-ins** – as AI models (once trained on open data platforms) can perform tasks that previously required proprietary tools, companies can migrate away from costly legacy software licensing. ENERGYai's vision explicitly includes *"forecasting a significant decline in traditional vendor lock-ins"* as AI agents take over tasks handled by big vendor suites.
- **Data Quality and Decision Velocity:** A subtler but important benefit is the improvement in data management and the speed of decision-making. With AI agents standardizing and validating data continuously, the confidence in data-driven decisions increases. Engineers spend less time reconciling conflicting data or waiting for reports – instead, they get near-real-time, AI-curated information. This improves the **velocity of decisions**, a critical advantage in fast-moving operational contexts (like drilling or managing a well intervention). In the ENERGYai trial, the "data preparation" time for generating a routine production report went from several days (when done via manual gathering from various systems) to instantaneous (with the AI portal generating a live dashboard). When decisions can be made faster, the whole operation becomes more responsive.

It should be noted that **not all benefits are "given" or guaranteed**, and real-world results depend on the quality of implementation and data. Some early attempts at AI have underperformed due to poor data quality or lack of user trust (discussed next).

In summary, agentic AI is already delivering concrete value in upstream operations: **faster analyses, smarter decisions, higher productivity, and lower risk**. The business case is compelling. Nonetheless, significant challenges and considerations remain on the path to full adoption, which we address in the following section.

Challenges and Considerations in Adopting Agentic AI

Despite its promise, implementing agentic AI in upstream O&G is not without challenges. Companies must navigate technical, organizational, and ethical hurdles to fully realize the benefits:

Data Quality and Standardization. AI systems are only as good as the data fed into them. Upstream data is notoriously messy – coming from different sources (geological, geophysical, operational), in varied formats, and often of uneven quality. Many operators have legacy databases with inconsistent data entries, missing metadata, or erroneous records. If an AI agent is trained or operates on bad data, it can produce flawed

recommendations (the classic "garbage in, garbage out" problem). Thus, a significant effort is needed to clean and standardize data **before and during** AI deployment. In the ENERGYai trial, a lot of initial work went into aggregating 50 years of data and harmonizing it under OSDU standards. Even then, the AI had to perform ongoing data QC – which, as noted, became a benefit as it improved data reliability by catching errors. Industry-wide, adoption of common data platforms (like OSDU) and data governance practices will be the key.

Model Validation and QA/QC of AI Outputs. Upstream decisions can involve high capital stakes and safety implications, so **quality assurance** is paramount. AI agents must be rigorously validated – both in development and in operation. This means testing the AI models on historical cases to see if they would have made sound recommendations, and continuously monitoring their outputs. In critical tasks like drilling control or safety monitoring, simulation testing of the AI is advisable (e.g. using digital twins of operations to see how the AI behaves in various scenarios). Many companies are instituting an "**AI QA/QC**" step in workflows: just as there is QA/QC for data, there is validation for AI suggestions. For example, if the AI picks a drilling trajectory, a drilling engineer might run a quick stress test with conventional models to double-check pressures and loads. Over time, as confidence grows, this duplication can reduce, but initially it's necessary to build trust. Moreover, AI models should be regularly updated or retrained as new data comes in – otherwise they might become stale or drift from reality. Having a clear version control and testing protocol for AI models (especially after continuous learning updates) is important to ensure they remain reliable. Some industries have introduced the concept of "Verified AI" or certified algorithms for certain tasks; upstream might see similar frameworks.

Talent and Skill Gaps. Deploying agentic AI requires a workforce that not only understands upstream engineering but also has literacy in AI and data science. There is a need for hybrid skillsets – petroleum engineers who can code, data scientists who grasp reservoir physics, etc. Many companies currently lack sufficient staff in this intersection. Bridging this gap might involve upskilling existing employees (through training in AI foundations, Python, ML basics, etc.) and hiring new talents (which is competitive given demand for AI skills in many industries). The cultural gap mentioned earlier is also a skills gap: seasoned domain experts might need support to become comfortable using AI tools. Conversely, data scientists might need training to appreciate oilfield realities so they don't develop models in a vacuum. Some firms are creating cross-functional "AI SWAT teams" for deployments – combining IT, data, and domain personnel. Another aspect is ensuring **knowledge transfer**: as AI takes over routine tasks that used to train junior staff, companies must find new ways to teach domain fundamentals (through simulations or rotating roles) so that human expertise still develops robustly. This is both a challenge and an opportunity to reimagine training in an AI-rich environment.

Change Management and Resistance. As discussed, not everyone will immediately trust an autonomous agent making decisions. Building trust in autonomous decision-making remains a **significant obstacle**. Workers may fear job displacement or loss of control, leading to resistance or underutilization of the AI tools. If a team doesn't *trust* the AI, they might ignore its recommendations (even good ones) or underuse its capabilities, negating potential gains. To overcome this, strong change management practices must be in place: involve end-users early, transparently communicate what the AI will do and won't do, and clarify that the goal is augmentation, not replacement. Early wins should be publicized internally – e.g., when the AI prevents an incident or finds additional reserves, share that success and credit both the technology and the team that used it. It often takes time and repeated positive experiences for skepticism to give way to trust. One strategy is to implement AI in advisory mode first (decision support), and only move to full autonomy in steps as confidence builds. The **human-in-the-loop** approach initially can help: e.g., the AI suggests, the human decides. As comfort increases, some decisions can be automated with human override.

Cybersecurity and System Risks. Introducing AI agents that potentially control operations opens new risk fronts, particularly in cybersecurity. If an autonomous agent can send commands to field equipment or influence setpoints, it becomes a target for malicious actors. Companies must ensure that the AI systems are secure – with proper authentication, encryption, and isolation from external networks. There's also the risk of the AI going awry due to a glitch or unforeseen scenario (the classic *rogue AI* fear, albeit far from Hollywood dramatization in real settings). Strong fail-safes are needed: e.g., if an AI agent issues a command outside of pre-approved parameters, the system should block it or require human confirmation. Ensuring that there is always a manual override or shutdown mechanism is critical for safety. Additionally, as AI uses large cloud-based models (like ENERGYai's 70B-parameter LLM[10]), data privacy and compliance must be addressed – sensitive reservoir data or well logs might be stored or processed in cloud environments, raising concerns about intellectual property or regulatory compliance (especially across borders). Governance policies around data handling by AI and audit trails for AI decisions will need to be established.

Performance in Novel Situations. AI agents excel when operating within the bounds of what they've been trained on. But the oilfield often presents novel situations (geological surprises, black swan events). A risk is that an AI might make a confident but wrong decision in an unprecedented scenario. Human experts, by contrast, might exercise caution or recognize when conditions are abnormal. Mitigating this involves expanding training data as much as possible and imbuing AI with uncertainty awareness (i.e., having the AI report when a situation falls outside its training distribution). Techniques like simulation and stress-testing can expose AI to edge cases. Nonetheless, companies should be mindful that **autonomy doesn't equal infallibility**. A prudent approach is using AI for what it's good at (data-heavy pattern recognition, fast response) and keeping humans in charge of strategic judgment calls, especially when venturing into the unknown. Over time, as AI proves itself, the boundary can shift, but it should be a deliberate process.

Regulatory and Ethical Concerns. The regulatory environment for AI in critical infrastructure is still nascent. Questions arise such as: Who is accountable if an AI makes a decision that leads to a spill or accident? The operator, the AI developer, or the machine itself? Regulators might start requiring validation or certification of AI systems used in safety-critical operations, like how aircraft autopilot software is certified. Companies should anticipate and even welcome some level of oversight to ensure AI is deployed responsibly. Ethically, the workforce impact must be managed – retraining programs and career transition support for roles that may be diminished by AI (e.g., certain data processing jobs) are part of being a responsible employer. Additionally, ensuring AI decisions align with environmental and safety ethics is crucial – for instance, an AI should not be optimizing production at the expense of safety margins or environmental regulations (unless explicitly directed otherwise). Clear guidelines and corporate policies need to frame the AI's objectives so that efficiency gains do not come at the cost of values or compliance.

In summary, while the **path to agentic AI** is incredibly promising, it requires careful handling of data readiness, validation, human factors, and safety nets. The companies that succeed will be those that *not only* invest in cutting-edge algorithms but also lay the groundwork in data infrastructure, cultivate an agile learning culture, and enforce rigorous oversight of AI operations. The challenges are surmountable with proper planning and a proactive approach – and the reward is a resilient, high-performing operation that stands to redefine industry benchmarks.

Conclusion

The upstream O&G industry stands on the cusp of a transformative AI-driven future. **Agentic AI**, with its autonomous and intelligent agents, promises to redefine how exploration and production activities are conducted – ushering in an era where machines collaborate with humans to achieve optimal outcomes in speed, efficiency, and safety. As we have discussed, this is more than just another step in automation; it is a fundamental disruption of traditional workflows and roles.

The experiences from ENERGYai's deployment and similar initiatives make one thing clear: **organizations that embrace this shift early will lead the pack.** By swiftly adopting AI-driven frameworks, companies can dramatically enhance their operational performance – reducing cycle times, improving recovery rates, cutting costs, and minimizing incidents. There is a competitive advantage in being an early mover: these technologies benefit from network effects (more data -> better AI -> more value) which means laggards will find it hard to catch up once leaders' AI agents become highly evolved and data-rich.

Moreover, empowering the workforce – especially the new generation of professionals – to harness AI will create a more innovative and adaptive organizational culture. Those firms that invest in their junior talent, allowing them to thrive alongside AI, will cultivate future leaders who are adept in both domain and digital expertise. The *digital natives*, with AI as their co-pilot, will drive the next leaps in efficiency and creative problem-solving in upstream. **Traditional giants** that remain reliant on legacy systems, siloed data, and old licensing models must recognize that clinging to the past is a risky strategy. Not only do they risk losing operational efficiency, but they may also struggle to attract new talent who expect modern, AI-augmented workplaces. In the worst case, such companies could face **obsolescence** if they cannot deliver competitive performance and agility in this new landscape.

That said, the journey must be undertaken responsibly. Success in the era of agentic AI will require balancing technological capability with governance. Companies should strive to become what ADNOC's CEO termed "*the world's most AI-enabled energy company*," but also one that ensures AI is *trustworthy*, safe, and aligned with its people and values. This includes continuing to uphold stringent QA/QC, maintaining human oversight where needed, and iterating on both the tech and the workflows as lessons are learned.

In conclusion, the future of upstream will be characterized by **hybrid intelligence** – an interplay of advanced AI agents and empowered humans working together to find and produce energy more effectively than ever before. The tools of the trade are changing: where once the geologist's key instruments were paper logs and seismic sections, now they are AI models and analytics dashboards. Where a drilling superintendent once relied on intuition and radio reports, now they have real-time AI advisors monitoring every parameter. It's a future where decisions are data-driven and often machine-suggested, but human insight and strategic thinking remain crucial. The companies that figure out this symbiosis will dominate the industry's next chapter.

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